Find the weights assigned to the three varieties by using Cramer's Rule.

6. A total of ₹ 8,500 was invested in three interest earning accounts. The interest rates were 2%, 3% and 6% if the total simple interest for one year was ₹380 and the amount invested at 6% was equal to the sum of the amounts in the other two accounts, then how much was invested in each account? (use Cramer's rule).

1.3 Transition Probability Matrices

1.3.1 Forecasting the succeeding state when the initial market share is given

One stage Transition Probability

The Occurrence of an event at a specified point in time, put the system in state S_n; if after the passage of one unit of time, another event occurs, that is the system moved from the state S_n to S_{n+1}. This movement is related to a probability distribution, there is a probability associated with each (move) transition from event S_n to S_{n+1}. This probability distribution is called one stage transition probability.

Transition Matrix

The transition Probabilities P_{jk} satisfy $P_{jk} > 0,\ \sum_k P_{jk} = 1$ for all j

These probabilities may be written in the matrix form

$$P = \begin{pmatrix} P_{11} & P_{12} & P_{13} & \cdots \\ P_{21} & P_{22} & P_{23} & \cdots \\ \cdots & \cdots & \cdots & \cdots \\ \cdots & \cdots & \cdots & \cdots \end{pmatrix}$$

This is called the transition probability matrix.

Example 1.25

Consider the matrix of transition probabilities of a product available in the market in two brands A and B.

$$\begin{array}{c} \\ A \\ B \end{array} \begin{array}{c} \begin{array}{cc} A & B \end{array} \\ \begin{pmatrix} 0.9 & 0.1 \\ 0.3 & 0.7 \end{pmatrix} \end{array}$$

Determine the market share of each brand in equilibrium position.

Solution:

Transition probability matrix

$$T = \begin{array}{c} \\ A \\ B \end{array} \begin{array}{c} \begin{array}{cc} A & B \end{array} \\ \begin{pmatrix} 0.9 & 0.1 \\ 0.3 & 0.7 \end{pmatrix} \end{array}$$

At equilibrium, $\begin{pmatrix} A & B \end{pmatrix}\ T = \begin{pmatrix} A & B \end{pmatrix}$ where $A + B = 1$

$$\begin{pmatrix} A & B \end{pmatrix}\begin{pmatrix} 0.9 & 0.1 \\ 0.3 & 0.7 \end{pmatrix} = \begin{pmatrix} A & B \end{pmatrix}$$

$$0.9A + 0.3B = A$$

$$0.9A + 0.3(1 - A) = A$$

$$0.9A - 0.3A + 0.3 = A$$

$$0.6A + 0.3 = A$$

$$0.4A = 0.3$$

$$A = \frac{0.3}{0.4} = \frac{3}{4}$$

$$B = 1 - \frac{3}{4} = \frac{1}{4}$$

Hence the market share of brand A is 75% and the market share of brand B is 25%

Example 1.26

Parithi is either sad (S) or happy (H) each day. If he is happy in one day, he is sad on the next day by four times out of five. If he is sad on one day, he is happy on the next day by two times out of three. Over a long run, what are the chances that Parithi is happy on any given day?

Solution:

The transition porbability matrix is

$$T = \begin{pmatrix} \frac{1}{3} & \frac{2}{3} \\ \frac{4}{5} & \frac{1}{5} \end{pmatrix}$$

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